

FamBench: Affinity Benchmarks Measure Interpolation Within Familiar Protein Families, Not Transfer to Novel Ones

Mehdi Yazdani Jahromi

myjahromi@gmail.com

July 22, 2026

Abstract

Co-folding models such as Boltz-2 and Nesso-1 report protein–ligand affinity correlations of Pearson ≈ 0.6 and are proposed as fast surrogates for free-energy perturbation. We show that this headline accuracy is largely *interpolation within protein families already represented in public structural data*, not prediction for novel targets. The central result is an interaction: frontier co-folders depend strongly on protein-family support while family-disjoint shallow controls do not. Defining the *family generalization gap* $G_m = r_m(S_f \geq 301) - r_m(S_f = 1)$ over matched-affinity strata, we measure $G_{\text{Nesso}} = 0.45$ (95% CI [0.27, 0.61]) and $G_{\text{Boltz}} = 0.62$ ([0.12, 1.01]), versus $G_{\text{RF}} = -0.01$ and $G_{\text{kNN}} = -0.03$ (CIs spanning zero) for a random-forest QSAR model and a ligand- k NN baseline under family-disjoint cross-validation. Family support predicts accuracy after controlling for ligand similarity, protein length, publication date, family affinity variance, measurement type, and affinity range (Nesso $t = -8.1$; Boltz $t = -3.7$); a two-dimensional analysis localizes the dependence to the *protein-family* axis, not ligand chemistry. On novel families the practical ranking inverts: a shallow family-disjoint random forest ($r = 0.41$) beats both co-folders (0.32, 0.31). A prospective, structurally novel post-cutoff target (OpenBind EV-A71 2A protease) supplies external confirmation: no co-folder reliably exceeds a molecular-weight baseline. Applying the same ligand-only control to the standard LBA30 benchmark—which no prior work reports—molecular weight alone ($r = 0.44$) matches or beats most of the published leaderboard. We call this effect *redundancy-driven inflation / training-familiarity dependence* rather than leakage, and release FamBench—a plug-in benchmark and dataset that measures accuracy across the full family-support curve, with family-disjoint controls and two-level bootstrap inference.

1 Introduction

Predicting protein–ligand binding affinity from structure is central to computational drug discovery. Co-folding models—AlphaFold3 [1], Boltz-1/2 [2, 3], Chai-1 [4], Nesso-1 [5], TerraBind [6]—increasingly predict affinity directly, with Boltz-2 approaching free-energy-perturbation accuracy [3]. These models are summarized by a single correlation on a held-out set. That number cannot separate a model that has learned binding physics from one that recognizes protein families it has seen many times in the PDB. The distinction is decisive: a medicinal chemistry program succeeds or fails on *novel* targets—those least represented in public structural and affinity data.

Prior work shows ligand-based and docking benchmarks reward memorization [7, 8], that affinity networks rely on train–test similarity rather than interaction physics [9, 10], and that re-splitting

lowers apparent accuracy [11]. What is missing is a quantitative, model-agnostic account of how co-folding affinity accuracy depends on *protein-family support*, and a benchmark that treats that support as the independent variable.

Central claim. Standard affinity benchmarks largely measure interpolation within protein families already represented in public structural and affinity data. FamBench measures transfer to unseen protein families. Under matched affinity distributions and family-disjoint controls, Nesso-1 and Boltz-2 improve sharply with family redundancy but collapse on singleton families, where a shallow random forest performs better. A prospective OpenBind target confirms that current co-folders may fail to reliably exceed simple physicochemical baselines.

Contributions.

1. **A model-class $\times$ family-support interaction.** We define the family generalization gap G_m and show it is large and significant for co-folders and indistinguishable from zero for family-disjoint controls (Sec. 3).
2. **Familiarity dependence, controlled.** Pre-cutoff family support predicts accuracy after ligand similarity, protein length, date, family affinity variance, measurement type, and affinity range; the dependence localizes to the protein-family axis, not ligand chemistry (Sec. 6.2). A family-mean predictor recovers most of the “accuracy” from family identity alone.
3. **Practical and prospective consequences.** On novel families the ranking inverts; on a novel post-cutoff target neither co-folder beats molecular weight (Secs. 6.3, 6.4).
4. **FamBench**, a plug-in benchmark and dataset with a two-dimensional (family-support $\times$ ligand-similarity) design, family-disjoint controls, and a two-level bootstrap reporting protocol (Sec. 4).

2 Related Work

Co-folding affinity models. AlphaFold2 [12] was extended to protein–ligand complexes by AlphaFold3 [1] and open reimplementations Boltz-1 [2], Chai-1 [4], RoseTTAFold-All-Atom [13]. Boltz-2 [3], Nesso-1 [5], and TerraBind [6] predict affinity directly, evaluated by overall correlation.

Scoring and QSAR baselines. Empirical (Vina [14], smina [15]) and learned (RF-Score [16], Gnina [17], AEV-PLIG [18]) functions are trained and tested on PDBbind [19] / CASF-2016 [20], with labels from BindingDB [21] and ChEMBL [22], against FEP+ [23].

Memorization and benchmark critiques. Wallach and Heifets [7] and Chen et al. [8] traced ligand-based/docking “success” to dataset bias; Volkov et al. [9] found ligand-only and protein-only models match full-complex models and nearest-neighbor baselines are already strong. Kanakala et al. [10] documented latent biases; MISATO [24] re-curated data.

Positioning against recent split-aware benchmarks. FamBench differs from three recent efforts by making *protein-family support* the primary axis on *frozen frontier co-folders*. Runs N’ Poses [25] studies pose recovery versus training similarity; FamBench studies affinity versus family support. NTAB [26] stratifies post-cutoff ChEMBL assays by *ligand* similarity; FamBench makes protein-family familiarity the main axis while controlling ligand novelty. Leak-Proof PDBBind [11]

constructs protected splits and *retrains* conventional scoring functions; FamBench audits frozen frontier co-folders across the full family-support curve and adds prospective evaluation. Nesso-1 [5] reports chemical-similarity analyses but explicitly lacks a disclosed target sequence/structure-similarity analysis; FamBench fills that missing evaluation. Sheridan [27] motivates the time-split view our OpenBind arm adopts.

3 Definitions

Family support. For a target in family f we define the pre-cutoff support S_f as the number of PDBbind structures in the same MMseqs2 [28] 30% sequence-identity cluster. Because our evaluation complexes are all deposited before the models’ 2021-09-30 training cutoff, S_f is a count of structures the models could have seen for that family. We bin S_f into $\{1, 2-5, 6-20, 21-80, 81-300, 301+\}$.

Matched strata. Within each family-support bin we compute Pearson r between predicted and measured affinity over a *matched* pK window $[4.5, 8.0]$, holding the label spread (and thus attainable correlation) approximately constant across bins.

Family generalization gap.

$$G_m = r_m(S_f \geq 301) - r_m(S_f = 1), \quad (1)$$

the change in accuracy from singleton to well-supported families. A positive G_m means accuracy is bought by family support. We estimate G_m with a *two-level bootstrap*: resample families with replacement, then resample ligands within each family, so the confidence interval reflects both family-level and ligand-level variability.

Controls. The honest counterpart to a frozen co-folder—whose test families are in its training window—is a model that *cannot* have seen the test family. We evaluate a random-forest QSAR model (RF-QSAR) and a ligand- k NN baseline under 5-fold *family-disjoint* cross-validation (GroupKFold on family id), so every prediction is on a held-out family.

4 The FamBench Benchmark

Redundancy set. 18,759 PDBbind-derived complexes with an exact $K_d/K_i/IC_{50}$ label (pK), each annotated with protein-family support S_f and ligand nearest-neighbor ECFP4 Tanimoto. A balanced 3,360-target “core” subset (560 per support bin) is provided for quick evaluation.

Temporal set. 649 compounds with measured K_d against one structurally novel, post-2021 target—the OpenBind EV-A71 2A protease [29] (CC0)—with molecular-weight, clogp, Gnina, smina, AEV-PLIG, and AQ-Affinity baselines shipped.

Two-dimensional design. FamBench stratifies by *protein-family support* $\times$ *ligand similarity*, separating two memorization channels: co-folders may exploit protein-family familiarity while ligand and models may exploit chemical similarity. Matching affinity distributions alone does not isolate these channels.

Table 1: Family generalization gap $G_m = r(S_f \geq 301) - r(S_f = 1)$ (matched- pK , two-level bootstrap 95% CI). Co-folders depend strongly on family support; family-disjoint and trivial controls do not.

Method	Class	$r(S_f=1)$	$r(S_f \geq 301)$	G_m	95% CI
Nesso-1	affinity co-folder	0.11	0.55	+0.45	[+0.27, +0.61]
Boltz-2	affinity co-folder	-0.03	0.59	+0.62	[+0.12, +1.01]
RF-QSAR	family-disjoint ML	0.24	0.23	-0.01	[-0.16, +0.16]
ligand- k NN	family-disjoint ML	0.24	0.21	-0.03	[-0.17, +0.11]
clogp	trivial	0.10	0.24	+0.15	[-0.00, +0.28]
mol. weight	trivial	0.02	0.01	-0.01	[-0.15, +0.15]

Reporting protocol. We recommend: equal-family weighting (not per-complex); mean within-stratum correlation rather than pooled-across-target correlation; Pearson, Spearman, centered RMSE, and pairwise ranking accuracy; a two-level (family then ligand) bootstrap; sensitivity to the clustering choice; and a baseline suite—family-mean (target identity), molecular weight, clogp, ligand- k NN, and RF-QSAR. The prospective OpenBind score is reported as *one external target*, not population-level proof.

5 Experimental Setup

We evaluate models in four classes. **Affinity co-folders:** Boltz-2 [3] (cuEquivariance kernels, ColabFold MSAs) and Nesso-1 [5] (ESM-2 embeddings [30]), from public weights at default settings; both output $\log_{10}(\text{IC}_{50})$, negated to a higher-is-stronger score. **Confidence-proxy co-folders:** Chai-1 [4] and ESMFold2, which predict structure and an interface confidence (ipTM) but *not* affinity; we score their ipTM as a proxy, clearly labeled. **Family-disjoint controls:** RF-QSAR (ECFP4 + amino-acid composition) and ligand- k NN, both under family-disjoint CV. **Trivial/identity baselines:** family-mean, molecular weight, clogp. AlphaFold3 has open code but request-only weights and no affinity head; it is provided as a pluggable stub, not evaluated.

6 Results

6.1 A model-class $\times$ family-support interaction

Table 1 reports the family generalization gap. The affinity co-folders have large, significant gaps ($G_{\text{Nesso}} = 0.45$, $G_{\text{Boltz}} = 0.62$; two-level bootstrap CIs exclude zero), while every family-disjoint and trivial control has $G \approx 0$ with a CI spanning zero. This is the central finding: the shallow controls are not weaker co-folders—they behave *differently* as family support changes. The full dose-response (Table 2, Fig. 1) shows the co-folders rising monotonically with support while the controls stay flat.

6.2 Familiarity dependence, isolated by controls

It survives every covariate. Regressing per-target absolute error on $\log_{10} S_f$ together with ligand similarity, protein length, publication year, family affinity variance, measurement type, and the affinity regime (pK , pK^2), the family-support coefficient remains large and significant: Nesso -0.155 ($t = -8.1$), Boltz -0.157 ($t = -3.7$). The dependence is not explained by any measured confound.

Table 2: Matched- pK Pearson r by family support. Co-folders rise steeply; controls are flat.

Method	protein-family support S_f					
	1	2–5	6–20	21–80	81–300	301+
Nesso-1	0.105	0.225	0.378	0.421	0.437	0.551
Boltz-2	−0.030	−0.007	−0.008	0.420	0.371	0.585
RF-QSAR	0.259	0.264	0.257	0.254	0.196	0.177
ligand- k NN	0.215	0.223	0.192	0.236	0.174	0.180
mol. weight	−0.032	0.009	−0.101	0.107	0.070	−0.010

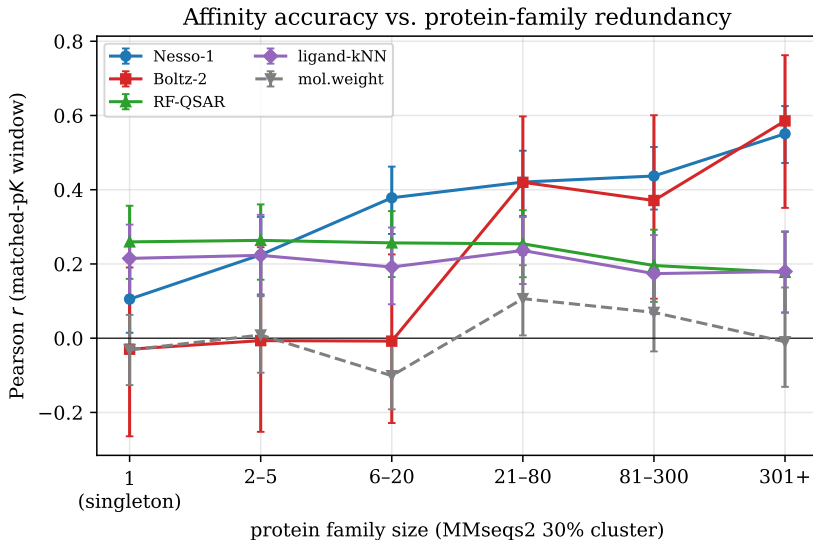


Figure 1: Affinity accuracy versus protein-family support (matched- pK , bootstrap 95% CIs). Co-folders rise steeply; family-disjoint RF and ligand- k NN are flat.

The channel is protein-family, not ligand chemistry. Table 3 stratifies Nesso-1 accuracy by family support $\times$ ligand similarity. Accuracy climbs down the family axis (0.1 $\rightarrow$ 0.5) but is flat across the ligand axis. The memorization channel for co-folders is protein-family familiarity.

Family identity alone recovers most of the “accuracy.” A predictor that outputs the mean pK of the target’s family scores Pearson **0.564** under random splitting (and 0 under family-disjoint splitting by construction)—close to the co-folders’ pooled 0.588. This memorization ceiling is corroborated by a modality ablation (Table 4, Fig. 2): a model given *only the protein*, blind to the ligand and so unable in principle to know the affinity, still reaches $r = 0.64$ (composition) or 0.65 (ESM-2 650M) under random splitting, collapsing to 0.25–0.33 under family-disjoint splitting.

6.3 On novel families the ranking inverts

The practical consequence is a rank inversion. On novel families ($S_f \leq 5$), the family-disjoint random forest ($r = 0.411$) *outperforms* both co-folders (Nesso 0.324, Boltz 0.308), and ligand- k NN (0.350) is competitive. A 10^3 -parameter model with no structure beats billion-parameter co-folders on exactly the regime that matters for a new target.

Table 3: Nesso-1 Pearson r by protein-family support $\times$ ligand similarity (matched-p K ; n in parentheses). The gradient runs along the family axis, not the ligand axis.

family support \ ligand	novel ligand	mid	similar ligand
novel (1–5)	+0.37 (170)	+0.10 (228)	+0.11 (325)
mid (6–80)	+0.44 (134)	+0.48 (253)	+0.35 (353)
redundant (81+)	+0.51 (152)	+0.53 (270)	+0.48 (262)

Table 4: Modality ablation and identity baseline (random forest; Pearson r). A protein-only model—blind to the ligand—nearly matches the full model under random splitting but collapses under family-disjoint splitting.

Features	Random split	Family-disjoint split	Gap
family-mean (identity)	0.564	≈ 0	—
ligand only (ECFP4)	0.676	0.503	0.17
protein only (composition)	0.637	0.248	0.39
protein only (ESM-2 650M)	0.653	0.330	0.32
full (ligand + protein)	0.740	0.456	0.28

6.4 Prospective confirmation on a novel target

The temporal set removes “family support” as a proxy: the EV-A71 2A protease post-dates the training cutoff. Table 5 reports within-target Spearman over 643 compounds. Molecular weight achieves $\rho = 0.469$; Nesso-1 barely exceeds it (0.486) and Boltz-2 underperforms it (0.395), as do Gnina, smina, AEV-PLIG, and AQ-Affinity. This is consistent with external caution: Nesso-1’s own report states its advantage over molecular weight on this target is not statistically convincing and that target sequence/structure similarity is unevaluated in its proprietary assays [5], and the OpenBind release itself flags molecular weight as a strong baseline for this campaign [29]. We report this as one external target, not population-level proof.

6.5 Confidence-proxy co-folders

Chai-1 and ESMFold2 predict structure and an interface confidence (ipTM), not affinity. Scoring their ipTM as an affinity proxy (Table 6), the signal is weaker than the affinity-head models, and the two proxies differ instructively. Chai-1 (overall Pearson 0.39, $n=598$) *itself* carries the family-support dependence—rising from 0.16 on novel families to 0.46 on redundant ones (family-support slope $t = -3.8$)—so the familiarity dependence is a property of the learned structural representation, not only of a trained affinity head. ESMFold2, by contrast (overall 0.16, $n=281$ via the hosted API), is a weak and noisy affinity proxy with *no* significant family-support dependence ($t = -0.7$); this is consistent with ESMFold2 being optimized for protein–protein interfaces rather than small molecules, so its protein–ligand ipTM is poorly calibrated for affinity. AlphaFold3 would enter this category but its weights are request-only.

6.6 The missing control on a standard benchmark: LBA30

Our thesis is not specific to the FamBench dataset. ATOM3D LBA [31] is the field’s standard structure-based affinity-generalization benchmark: PDBbind-refined complexes split by 30% protein sequence identity (“LBA30”; 3507 train / 490 test), on which recent geometric deep nets report large gains—IPBind [32] reports Pearson 0.732, a claimed 19.6% relative improvement. Yet no pub-

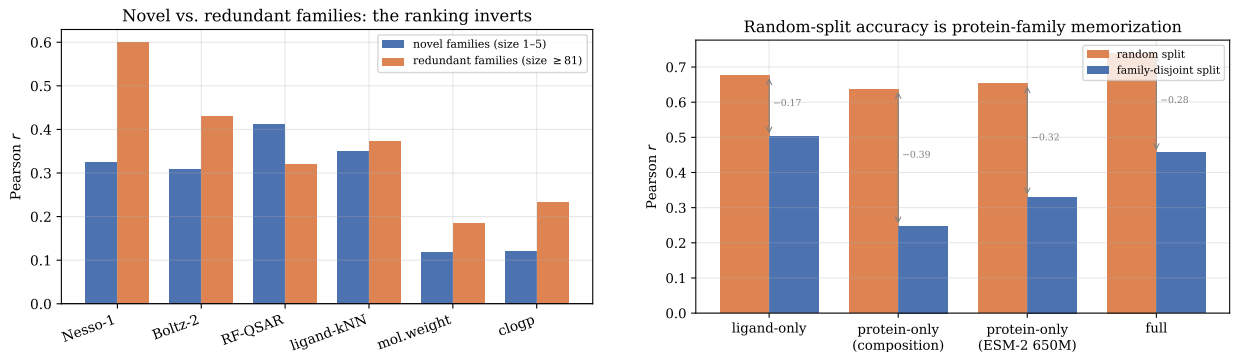


Figure 2: **Left:** novel- vs. redundant-family Pearson r ; the co-folders’ advantage is confined to redundant families. **Right:** random-split accuracy is protein-family memorization—a protein-only model collapses under family-disjoint splitting.

Table 5: Novel-target ranking (OpenBind EV-A71 2A protease, 643 compounds). Spearman ρ , and ρ after removing molecular weight.

Method	Spearman ρ	ρ beyond mol. weight
Nesso-1	0.486	0.233
molecular weight	0.469	—
Gnina	0.431	−0.082
Boltz-2	0.395	0.112
smna	0.229	−0.009
AEV-PLIG	0.228	−0.143
clogp	0.163	−0.057
AQ-Affinity	0.134	−0.019

lished LBA30 result reports a ligand-only, molecular-weight, or QSAR control; the field compares only against other structure/sequence deep nets. We supply that control (Table 7). A ligand-only random forest (ECFP4, no protein) scores 0.457 and molecular weight *alone* scores 0.438.

Two conclusions follow, and we state both fairly. First, the top methods genuinely learn structural signal: IPBind’s 0.732 clears the ligand-only ceiling by ≈ 0.28 , so it is not ligand memorization. Second, and more consequential for the field, molecular weight alone matches or exceeds much of the published LBA30 leaderboard—above ENN (0.389), on par with DeepDTA (0.472), and within 0.10 of 3DCNN (0.550) and GNN (0.545). Small reported margins over these methods were never contextualized against a trivial ligand descriptor. The same control also reconciles LBA30 with FamBench: the 30%-*sequence* split is a far weaker generalization test than singleton-family stratification—ligand-only reaches 0.46 on LBA30 but collapses to 0.31–0.41 on FamBench novel families—so 0.73 (LBA30) and 0.31 (FamBench novel) do not contradict; they measure different stringencies.

A related benchmark, CORDIAL [33], is explicitly designed to exclude fold/family identity and validated leave-superfamily-out; it too reports no ligand-only or trivial baseline. The absence of such controls across the generalization-benchmark literature is precisely the gap FamBench institutionalizes.

Table 6: Confidence-proxy co-folders (interface ipTM scored as an affinity proxy). Weaker than affinity heads, but the well-powered Chai-1 ipTM shows the same family-support dependence.

Method	n	overall r	novel-fam. r	redundant-fam. r
Chai-1 (ipTM)	598	0.394	0.155	0.461
ESMFold2 (ipTM)	281	0.163	0.104	-0.00

Table 7: LBA30 (test $n=453$), Pearson r on pK . Structure/sequence deep-net numbers are from the literature [31, 32]; the three ligand-only controls (bold) are ours and have not previously been reported. Molecular weight alone matches or beats most of the leaderboard; only the recent top methods clearly exceed it.

Structure/sequence models	r	Ligand-only controls (ours)	r
IPBind	0.732	ligand-only RF (ECFP4)	0.457
EHIGN	0.612	molecular weight	0.438
GIGN	0.586	ligand-kNN	0.429
3DCNN	0.550	clogp	0.212
GNN	0.545		
DeepDTA	0.472		
ENN	0.389		

7 Discussion

Why not “leakage.” Our evidence establishes *training-familiarity dependence* and *redundancy-driven inflation*: accuracy scales with pre-cutoff family support after controlling for confounds, and matched family-disjoint models do not share the dependence. We deliberately avoid the word “leakage,” which should be reserved for evaluations where related families demonstrably occur across a claimed train/test boundary. We cannot reconstruct the exact proprietary affinity training sets of the co-folders, so we make the weaker, defensible claim: public affinity benchmarks are dominated by families the models are familiar with, and that familiarity—not binding physics—carries most of the reported correlation.

Consequences. Co-folders are the most accurate methods on well-supported families; the problem is the standard practice of reporting one overall correlation, which conflates interpolation within familiar families with transfer to novel ones. The honest number for a new target is the novel-family value (≈ 0.1 – 0.3), and a cheap family-disjoint QSAR is a strong, rarely-reported prospective baseline.

Limitations. The redundancy set is pre-cutoff PDBbind; S_f is a within-corpus support proxy, not the models’ exact training counts. The temporal set is a single target with a congeneric series. We evaluate frozen public weights and do not retrain on de-leaked data. Family support at 30% identity is one axis; Pfam- and pocket-based clustering are natural sensitivity analyses. The confidence-proxy results use ipTM, not a trained affinity head.

8 Reproducibility and Artifact

FamBench is an installable package (`pip install fambench`) with a Hugging Face dataset. Scoring a model requires wrapping it as `predict(sequence, smiles)→float` and calling `run_redundancy`

/ `run_temporal`; a CLI scores a predictions CSV. All metrics, controls (G_m , matched- pK , multivariate regression, two-level bootstrap), baselines, and the dataset pipeline are included; every number here is reproduced by the shipped harness from released predictions.

9 Conclusion

Standard affinity benchmarks largely measure interpolation within familiar protein families. FamBench re-centers evaluation on transfer to unseen families: the family generalization gap separates frontier co-folders (large, positive) from family-disjoint controls (zero), a prospective novel target confirms neither co-folder reliably beats molecular weight, and the fix is to report the gap and the novel-family number rather than a single correlation.

Data and code. <https://github.com/DeepBioScientific/fambench>; <https://huggingface.co/datasets/DeepBioScientific/fambench>.

References

- [1] Josh Abramson, Jonas Adler, Jack Dunger, Richard Evans, Tim Green, Alexander Pritzel, Olaf Ronneberger, Lindsay Willmore, Andrew J. Ballard, Joshua Bambrick, et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, 630(8016):493–500, 2024. doi: 10.1038/s41586-024-07487-w.
- [2] Jeremy Wohlwend, Gabriele Corso, Saro Passaro, Mateo Reveiz, Ken Leidal, Wojtek Swiderski, Tally Portnoi, Itamar Chinn, Jacob Silterra, Tommi Jaakkola, and Regina Barzilay. Boltz-1: Democratizing biomolecular interaction modeling. *bioRxiv*, 2024. doi: 10.1101/2024.11.19.624167.
- [3] Saro Passaro, Gabriele Corso, Jeremy Wohlwend, Mateo Reveiz, Stephan Thaler, Vignesh Ram Somnath, Noah Getz, Tally Portnoi, Julien Roy, Hannes Stark, et al. Boltz-2: Towards accurate and efficient binding affinity prediction. *bioRxiv*, 2025. doi: 10.1101/2025.06.14.659707.
- [4] Chai Discovery. Chai-1: Decoding the molecular interactions of life. *bioRxiv*, 2024. doi: 10.1101/2024.10.10.615955.
- [5] Nikhil Shenoy, David Errington, Emmanuel Bengio, Kacper Kapuśniak, Kerstin Klaeser, Yui Tik Pang, Vladimir Radenkovic, Prudencio Tossou, Therence Bois, Andrew Wedlake, and Francesco Di Giovanni. Nesso-1: Accelerating open-source binding affinity predictions. Technical report, Valence Labs / Recursion, 2026. Technical report. <https://github.com/recursionpharma/nesso>.
- [6] Terray Therapeutics. TerraBind: Fast and accurate binding affinity prediction through coarse structural representations. *arXiv preprint arXiv:2602.07735*, 2026.
- [7] Izhar Wallach and Abraham Heifets. Most ligand-based classification benchmarks reward memorization rather than generalization. *Journal of Chemical Information and Modeling*, 58(5):916–932, 2018. doi: 10.1021/acs.jcim.7b00403.
- [8] Lieyang Chen, Anthony Cruz, Steven Ramsey, Callum J. Dickson, José S. Duca, Viktor Hornak, David R. Koes, and Tom Kurtzman. Hidden bias in the DUD-E dataset leads to misleading performance of deep learning in structure-based virtual screening. *PLOS ONE*, 14(8):e0220113, 2019. doi: 10.1371/journal.pone.0220113.
- [9] Mikhail Volkov, Joseph-André Turk, Nicolas Drizard, Nicolas Martin, Brice Hoffmann, Yann Gaston-Mathé, and Didier Rognan. On the frustration to predict binding affinities from protein–ligand structures with deep neural networks. *Journal of Medicinal Chemistry*, 65(11):7946–7958, 2022. doi: 10.1021/acs.jmedchem.2c00487.

- [10] Ganesh Chandan Kanakala, Rishal Aggarwal, Divya Nayar, and U. Deva Priyakumar. Latent biases in machine learning models for predicting binding affinities using popular data sets. *ACS Omega*, 8(2): 2389–2397, 2023. doi: 10.1021/acsomega.2c06781.
- [11] Jie Li, Xingyi Guan, Oufan Zhang, Kunyang Sun, Yingze Wang, Dorian Bagni, and Teresa Head-Gordon. Leak proof PDBBind: A reorganized dataset of protein–ligand complexes for more generalizable binding affinity prediction. *arXiv preprint arXiv:2308.09639*, 2023.
- [12] John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Židek, Anna Potapenko, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596(7873):583–589, 2021. doi: 10.1038/s41586-021-03819-2.
- [13] Rohith Krishna, Jue Wang, Woody Ahern, Pascal Sturmfels, Preetham Venkatesh, Indrek Kalvet, Gyu Rie Lee, Felix S. Morey-Burrows, Ivan Anishchenko, Ian R. Humphreys, et al. Generalized biomolecular modeling and design with RoseTTAFold All-Atom. *Science*, 384(6693):eadl2528, 2024. doi: 10.1126/science.adl2528.
- [14] Oleg Trott and Arthur J. Olson. AutoDock Vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of Computational Chemistry*, 31(2):455–461, 2010. doi: 10.1002/jcc.21334.
- [15] David Ryan Koes, Matthew P. Baumgartner, and Carlos J. Camacho. Lessons learned in empirical scoring with smina from the CSAR 2011 benchmarking exercise. *Journal of Chemical Information and Modeling*, 53(8):1893–1904, 2013. doi: 10.1021/ci300604z.
- [16] Pedro J. Ballester and John B. O. Mitchell. A machine learning approach to predicting protein–ligand binding affinity with applications to molecular docking. *Bioinformatics*, 26(9):1169–1175, 2010. doi: 10.1093/bioinformatics/btq112.
- [17] Andrew T. McNutt, Paul Francoeur, Rishal Aggarwal, Tomohide Masuda, Rocco Meli, Matthew Ragoza, Jocelyn Sunseri, and David Ryan Koes. GNINA 1.0: molecular docking with deep learning. *Journal of Cheminformatics*, 13(1):43, 2021. doi: 10.1186/s13321-021-00522-2.
- [18] Ísak Valsson, Matthew T. Warren, Charlotte M. Deane, Aniket Magarkar, and Philip C. Biggin. Narrowing the gap between machine learning scoring functions and free energy perturbation using augmented data. *Communications Chemistry*, 8:41, 2025. doi: 10.1038/s42004-025-01428-y.
- [19] Zhihai Liu, Minyi Su, Li Han, Jie Liu, Qifan Yang, Yan Li, and Renxiao Wang. Forging the basis for developing protein–ligand interaction scoring functions. *Accounts of Chemical Research*, 50(2):302–309, 2017. doi: 10.1021/acs.accounts.6b00491.
- [20] Minyi Su, Qifan Yang, Yu Du, Guoqin Feng, Zhihai Liu, Yan Li, and Renxiao Wang. Comparative assessment of scoring functions: The CASF-2016 update. *Journal of Chemical Information and Modeling*, 59(2):895–913, 2019. doi: 10.1021/acs.jcim.8b00545.
- [21] Michael K. Gilson, Tiqing Liu, Michael Baitaluk, George Nicola, Linda Hwang, and Jenny Chong. BindingDB in 2015: A public database for medicinal chemistry, computational chemistry and systems pharmacology. *Nucleic Acids Research*, 44(D1):D1045–D1053, 2016. doi: 10.1093/nar/gkv1072.
- [22] David Mendez, Anna Gaulton, A. Patrícia Bento, Jon Chambers, Marleen De Veij, Eloy Félix, et al. ChEMBL: towards direct deposition of bioassay data. *Nucleic Acids Research*, 47(D1):D930–D940, 2019. doi: 10.1093/nar/gky1075.
- [23] Lingle Wang, Yujie Wu, Yuqing Deng, Byungchan Kim, Levi Pierce, Goran Krilov, Dmitry Lupyan, Shaughnessy Robinson, Markus K. Dahlgren, Jeremy Greenwood, et al. Accurate and reliable prediction of relative ligand binding potency in prospective drug discovery by way of a modern free-energy calculation protocol and force field. *Journal of the American Chemical Society*, 137(7):2695–2703, 2015. doi: 10.1021/ja512751q.

- [24] Till Siebenmorgen, Filipe Menezes, Sabrina Benassou, Erinc Merdivan, Kieran Didi, André Santos Dias Mourão, Radosław Kitel, Pietro Liò, Stefan Kesselheim, Marie Piraud, Fabian J. Theis, Michael Sattler, and Grzegorz M. Popowicz. MISATO: machine learning dataset of protein–ligand complexes for structure-based drug discovery. *Nature Computational Science*, 4:367–378, 2024. doi: 10.1038/s43588-024-00627-2.
- [25] Runs N’ Poses contributors. Runs N’ Poses: Benchmarking pose prediction against training-set similarity, 2025. Benchmark referenced in Boltz-2 and Nesso-1 evaluations. VERIFY citation details.
- [26] NTAB contributors. NTAB: Novel-target affinity benchmark stratifying post-cutoff ChEMBL assays by ligand similarity, 2025. VERIFY citation details before submission.
- [27] Robert P. Sheridan. Time-split cross-validation as a method for estimating the goodness of prospective prediction. *Journal of Chemical Information and Modeling*, 53(4):783–790, 2013. doi: 10.1021/ci400084k.
- [28] Martin Steinegger and Johannes Söding. MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11):1026–1028, 2017. doi: 10.1038/nbt.3988.
- [29] OpenBind Consortium. OpenBind structure–affinity data release: Enterovirus a71 2a protease. Zenodo, version v1, 2026. Benchmark: <https://github.com/OpenBind-Consortium/A71EV2A-benchmark>; <https://openbind.uk>.
- [30] Alexander Rives, Joshua Meier, Tom Sercu, Siddharth Goyal, Zeming Lin, Jason Liu, Demi Guo, Myle Ott, C. Lawrence Zitnick, Jerry Ma, and Rob Fergus. Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. *Proceedings of the National Academy of Sciences*, 118(15):e2016239118, 2021. doi: 10.1073/pnas.2016239118.
- [31] Raphael J. L. Townshend, Martin Vögele, Patricia Suriana, Alexander Derry, Alexander Powers, Yianni Laloudakis, Sidhika Balachandar, Bowen Jing, Brandon Anderson, Stephan Eismann, Risi Kondor, Russ B. Altman, and Ron O. Dror. ATOM3D: Tasks on molecules in three dimensions. *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2021. arXiv:2012.04035.
- [32] Krinos Li, Xianglu Xiao, Zijun Zhong, and Guang Yang. Accurate and generalizable protein–ligand binding affinity prediction with geometric deep learning. *arXiv preprint arXiv:2504.16261*, 2025.
- [33] Benjamin P. Brown. A generalizable deep learning framework for structure-based protein–ligand affinity ranking. *Proceedings of the National Academy of Sciences*, 2025. doi: 10.1073/pnas.2508998122.